自动驾驶高精地图技术简介

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小鹏

扫码观看公开课

提 纲

- 自动驾驶是否需要高精地图
- 什么是高精地图
- 高精地图的生产
- 高精地图主要厂商

自动驾驶

an old concept becomes new again

1956 – AI

- Perceptron
- NN
- DL
- ...

2016~ AI 爆发

1961 – Stanford Cart

- NavLab
- Sandstorm
- Google Prius
- ...

2020~ 单车智能

自动驾驶等级定义

Level1
辅助驾驶
System takes care of one motor function
e.g. ACC, LCC
Level2
部分自动驾驶
System takes care of one motor function
e.g. ACC + LCC
Level3
条件自动驾驶
System can fully drive the vehicle autonomously
somewhere + sometime
Level4
高度自动驾驶
System can fully drive the vehicle autonomously
somewhere + anytime
Level5
完全自动驾驶
System can fully drive the vehicle autonomously
anywhere + anytime

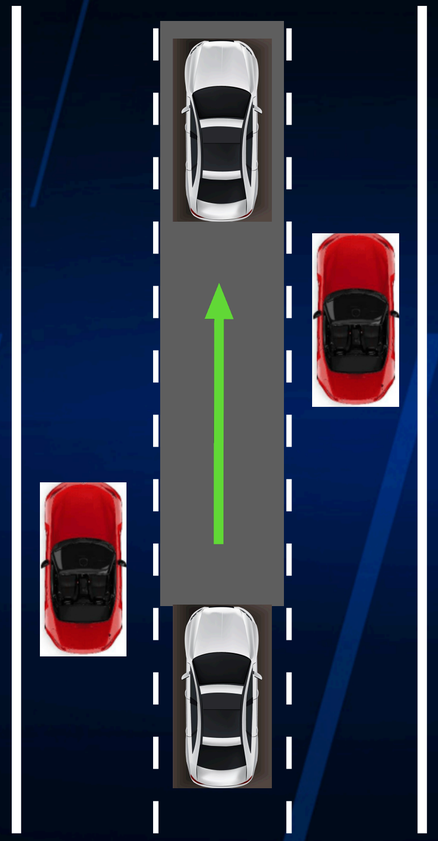
自动驾驶基本要素

感知 × 定位 × 规划 × 控制

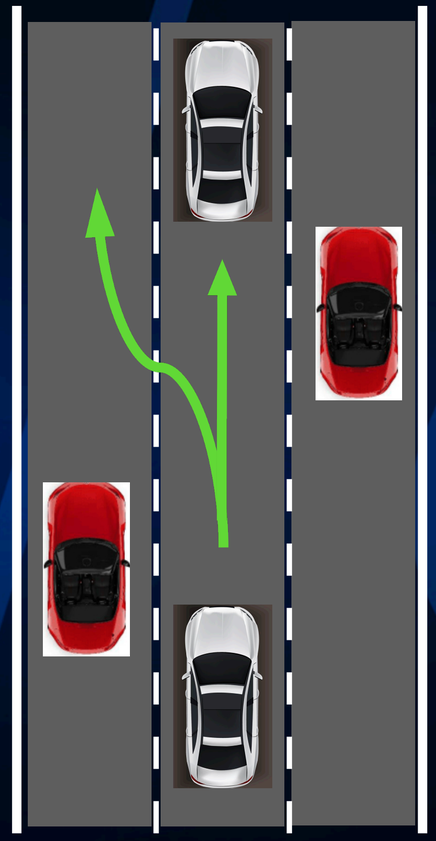
周围环境是什么样子　我在哪　应该怎么开　走起

典型行车场景

L2



L3



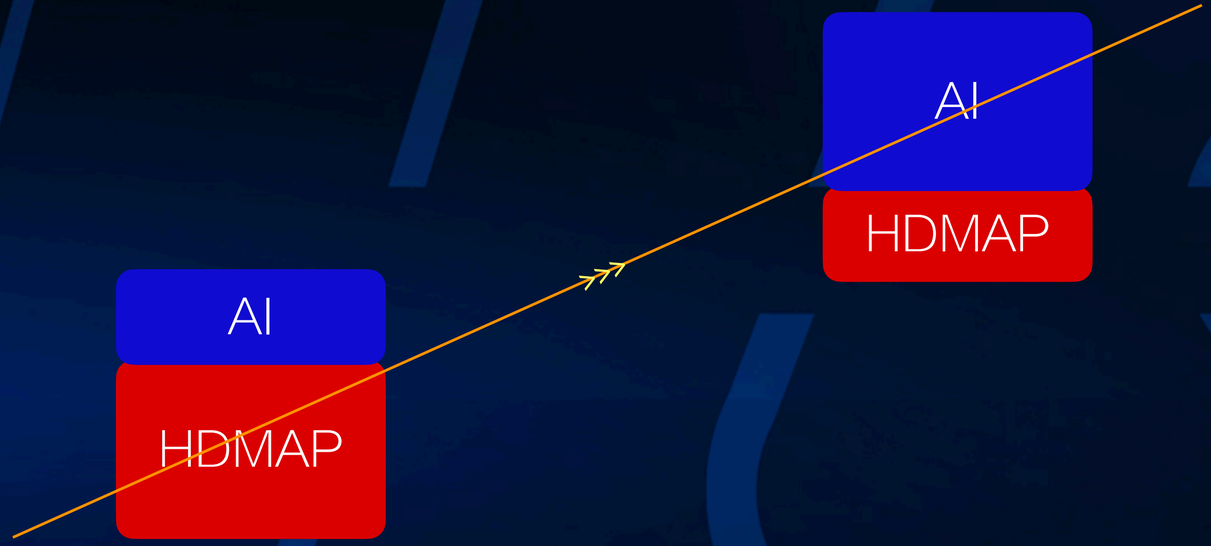
自动驾驶是否需要高精地图

- 人驾驶不需要高精地图
  - 熟悉道路: 人眼感知
  - 不熟悉道路: 道路级导航地图 + 人眼感知

- 理论上车载感知系统能获取所有道路信息:
  - 可行驶路径
  - 车道优先级
  - 车道线语义属性与通行逻辑
  - 红绿灯与车道通行关系
  - …

自动驾驶是否需要高精地图

- 传感器性能缺陷:
  - 视觉
  - Lidar
  - 弱势场景
    - 雨雾
    - 大雪
    - …
- AI能力不够:
  - 归纳能力强
  - 演绎能力弱
- 高精地图:
  - 将有辅助作用, 短期不变的道路先验信息存起来



先验信息差异化

- 储存哪些先验信息？
  - 路面元素: 车道线, 箭头, 停止线, 让行标志…
  - 高出路面元素: 交通信号灯, 标志牌, 龙门架, 杆子…
  - …
- 这些先验信息怎么组织和表达？
  - 点云
  - 体素
  - 稀疏特征点
  - 语义级
  - …

业内没有统一标准

自动驾驶核心能力建设

	L2	L3
视觉感知	前向感知为主 移动物体: 2dBB, 种类, 行人 路面 / 静止物体: 车道线 可达空间 / 未标注物体 / 异常情景	360度感知 移动物体: 3dBB, 转向灯, 横 / 纵向位置, 大小, 行人状态 路面 / 静止物体: 路牌, 路面标志, 交通灯, 几何+语义, 减速带…
定位/地图融合	相对定位(DR)	厘米级定位(横向), 地图仲裁
传感器融合	前向融合, 精准	360度, 可达空间
高精地图	–	高精地图检测, 关联
行为 / 运动预测	–	基于规则 / 运动轨迹的预测
行为规划	–	基于规则的规划; 路权谈判
运动规划 / 控制	轨迹跟踪	运动轨迹规划生成, 选择; 轨迹跟踪

高精地图可以干什么

我在哪里?

定位辅助

- 高精地图以定位的地图元素及专门的定位图层为基础，通过与本地车辆传感器实时数据匹配，获取最新的定位信息

我前方有什么?

感知辅助

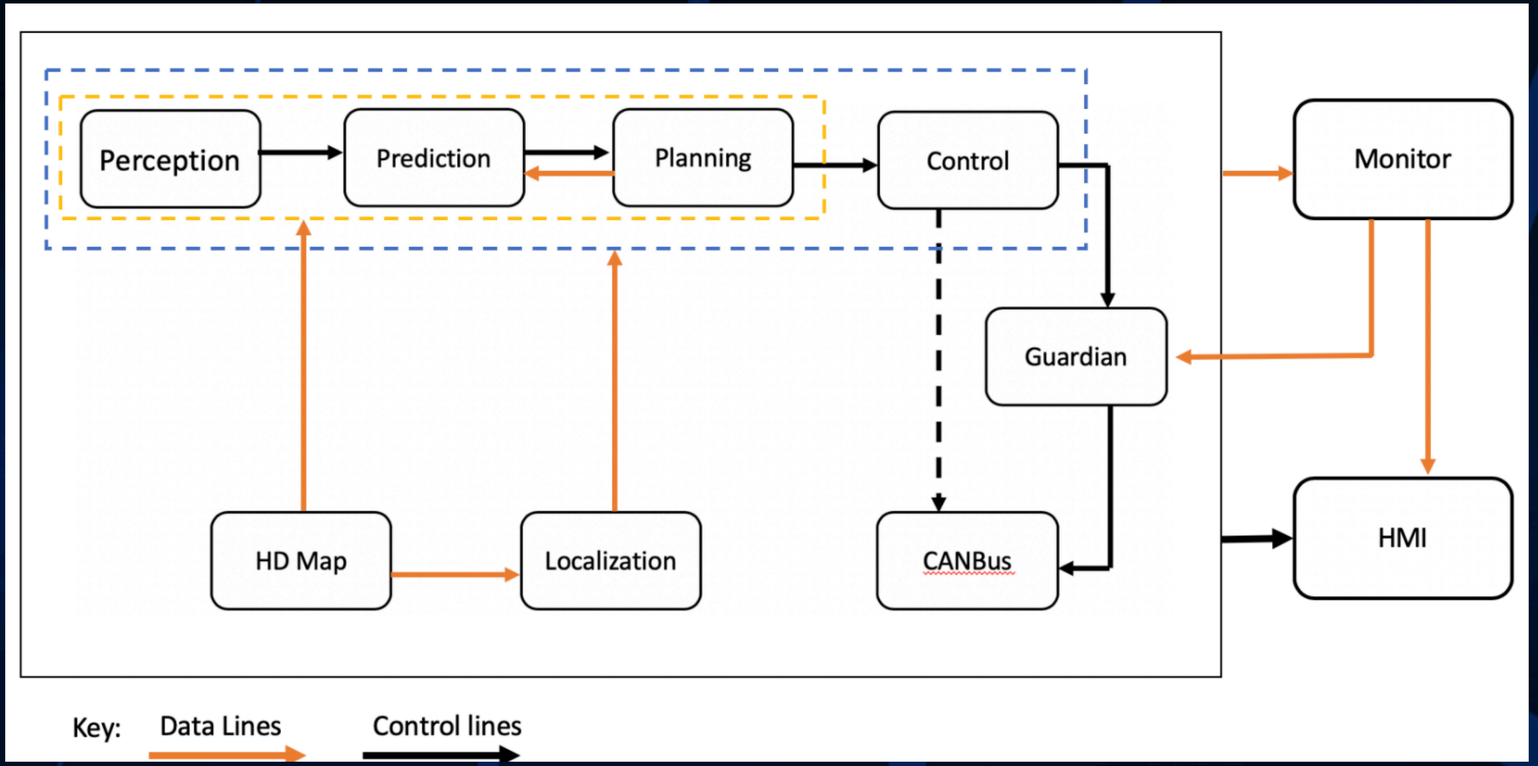
- 更远的距离
- 更稳定的性能
- 更高精度的采集
- 更全面的交通信息

我该怎么走?

决策辅助

- 通过高精地图的预知信息，自动驾驶辅助系统可以提前做出预先减速，避让，换道等更安全的路径规划

百度Apollo5.5软件架构



https://github.com/ApolloAuto/apollo

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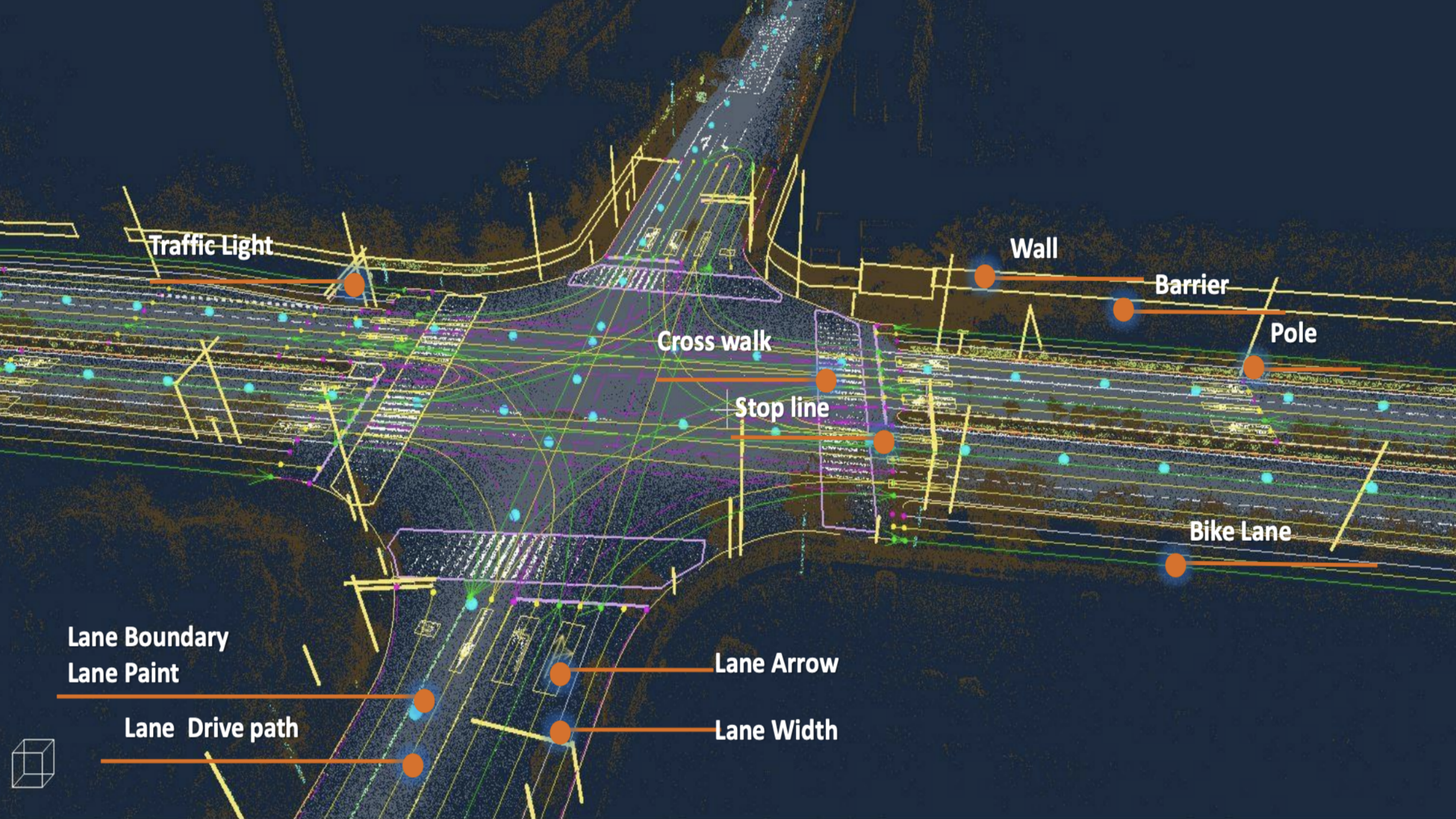
什么是高精地图

	SD MAP	HD MAP
数据主体	道路	车道
数据精度	道路级分辨率 (5m~10m)	车道级分辨率 (分米级)
服务对象	人	机器
主要用途	导航 + 搜索	定位辅助 / 感知辅助 / 决策辅助
系统类别	娱乐信息系统	车载安全系统
地图内容	道路形状, POI	更丰富, 车道线几何 / 逻辑, 道路设施…


Traffic Light
Wall
Barrier
Pole
Cross walk
Stop line
Bike Lane
Lane Boundary
Lane Paint
Lane Arrow
Lane Drive path
Lane Width

高精地图的内容

几何

+

逻辑

- 语义
- 拓扑

高精地图数据模型

数据模型

- 空间信息:
  - 车道模型
  - 道路设施
- 语义信息:
  - 环境
  - 物体
- 时间信息:
  - 潮汐街道
  - 红绿灯
  - 电子标识盘

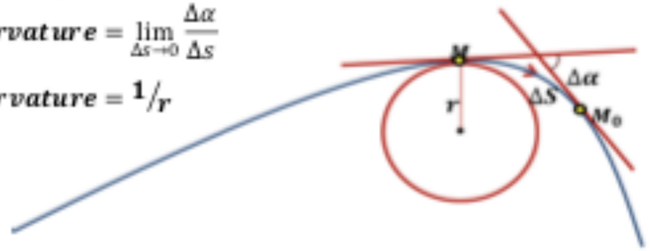
高精地图数学属性

曲率 Curvature

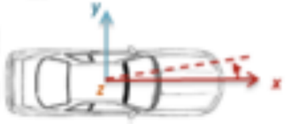
$M \to M_0$

$Curvature = \lim_{\Delta s \to 0} \frac{\Delta \alpha}{\Delta s}$

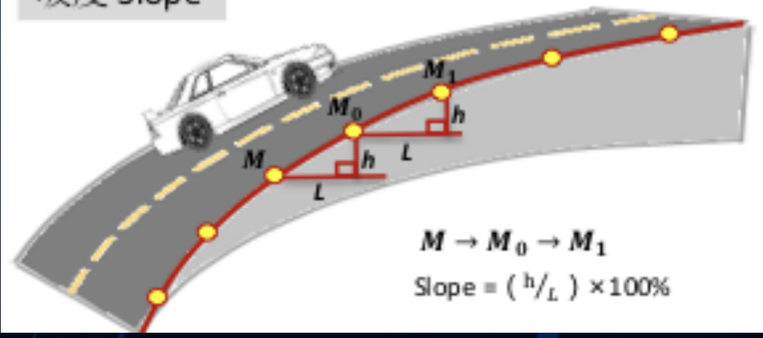
$Curvature = {}^{1}/_{r}$



航向 Heading



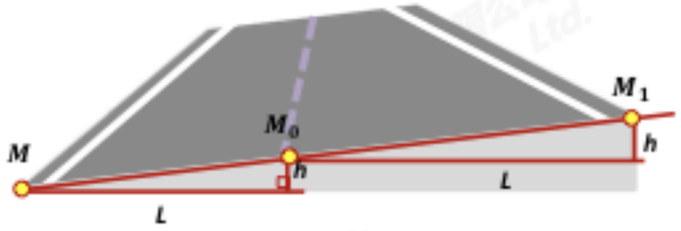
坡度 Slope



$M \to M_0 \to M_1$

Slope = $({}^{h}/_{L}) \times 100\%$

横坡 Super elevation

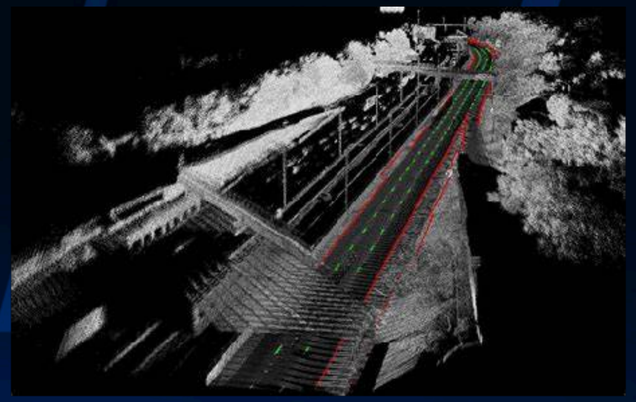


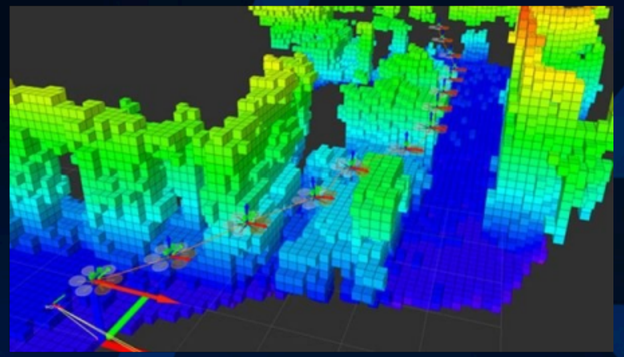
$M \to M_0 \to M_1$

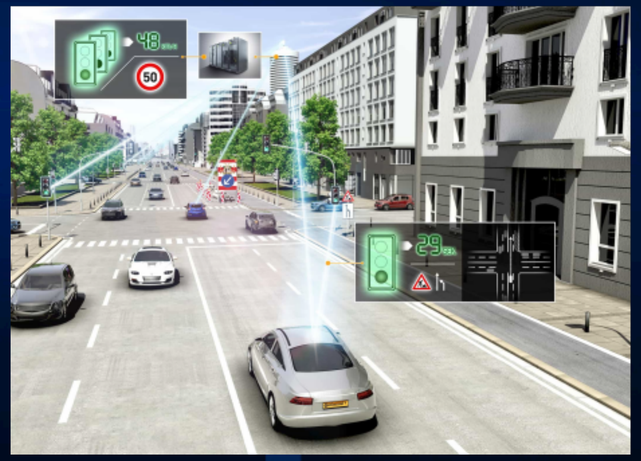
Super elevation = $({}^{h}/_{L}) \times 100\%$

高精地图中的定位图层

- 定位层的作用
  - 在地图中提供车辆一些绝对或者相对位置定位的信息
- 定位图层的内容:
  - Landmark
  - 激光点云或者体素
  - 稀疏特征点或者语义点
  - 路面特征
  - 其它







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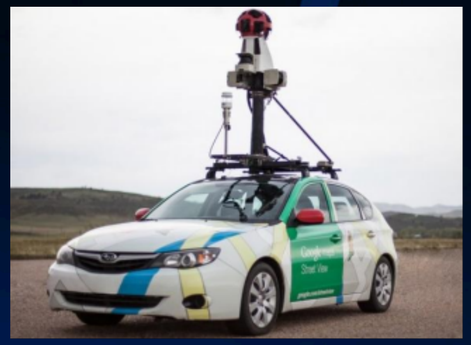
高精地图采集测绘数据的方式

- 专用采集车
  - Google, Here等图商
  - 价格昂贵，数据精准
  - 生产周期长

- 众包采集:
  - Tesla, MobileEye等新地图公司
  - 数据精度低
  - 算法复杂度高
  - 生产周期短



高精地图制作方式

集中制图

高德 AutoNavi 地图导航专家

Baidu 百度 百度一下 你就知道

TOMTOM

NAVINFO 四维图新

WAYMO

众包制图

MOBILEYE An Intel Company

momenta

lvl5

宽·登 KUANDENG

DEEPMAP

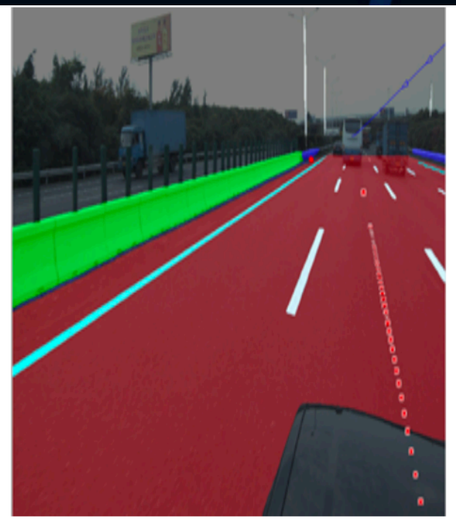
集中制图数据采集

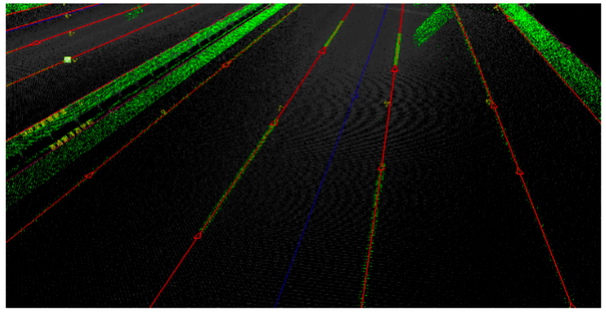
- 传感器
  - Lidar：采集点云数据，用以提供几何信息（路面宽度，红绿灯高度）。
    - 采集点云数据
  - 摄像头：
    - 优势：像素信息丰富，用来感知车道线，路面的标识等
    - 缺点： 没有深度信息。
  - GNSS：车辆位置信息
    - 优点：没有累计误差
    - 缺点：存在局部误差，不光滑。GNSS 容易受环境影响，不稳定。
  - IMU： 用来捕捉车辆的角度和加速度信息
    - 优点：不受环境影响，鲁棒。
    - 缺点：需要校订，有累计误差。

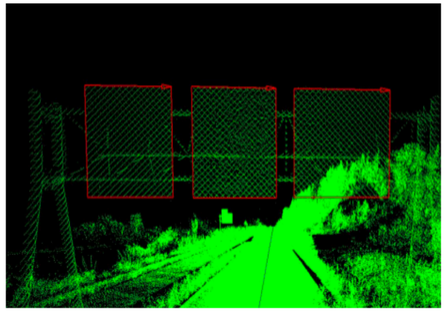
数据处理

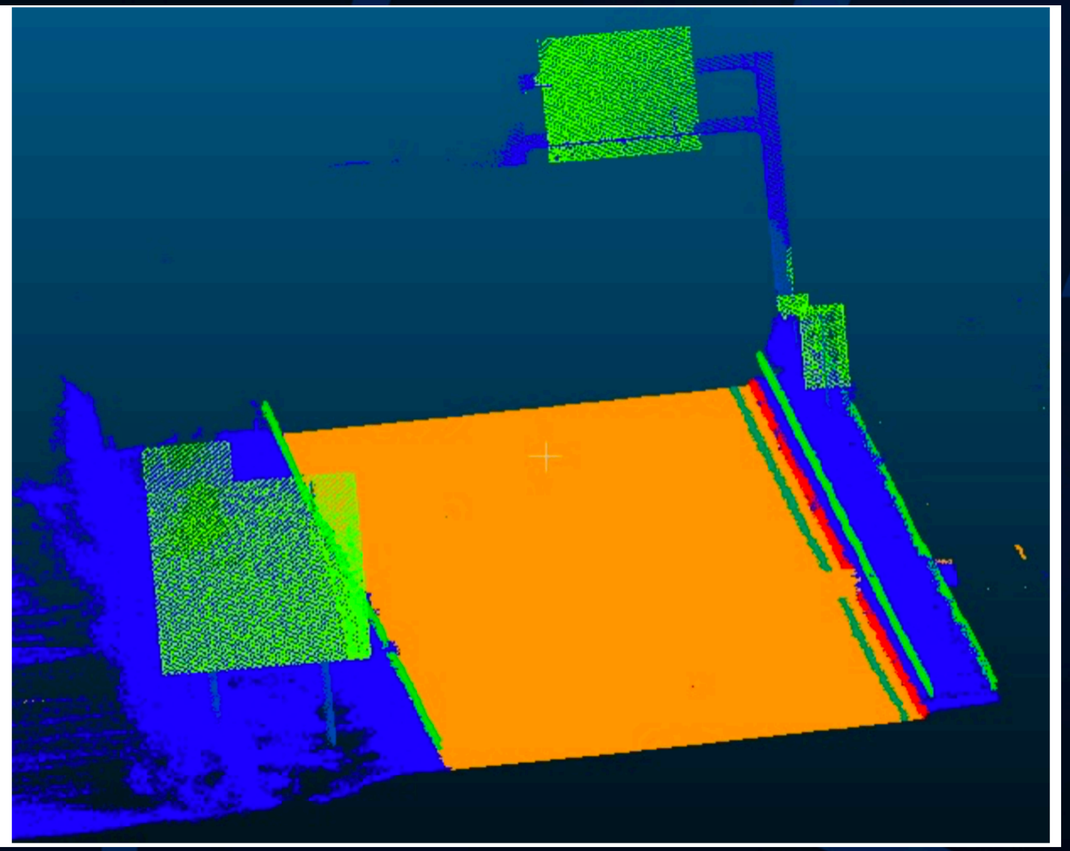
数据处理： 半自动化

- 点云数据清洗：
- 点云数据拼接：
- 图片拼接，与点云对齐
- 交通标识物检测
- 数据矢量化 和 检索

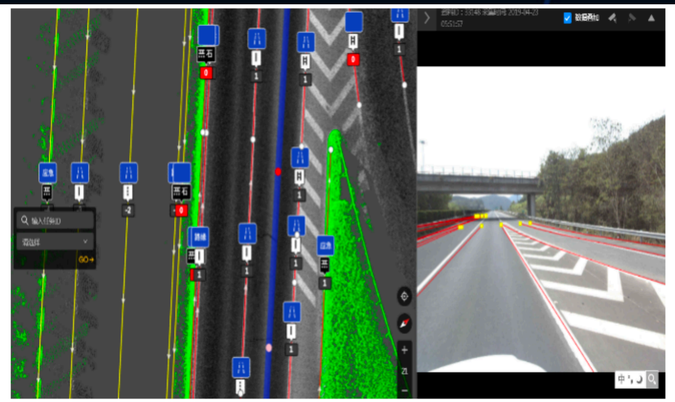


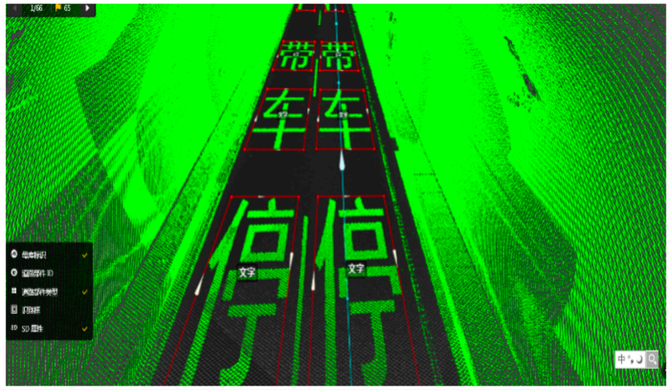


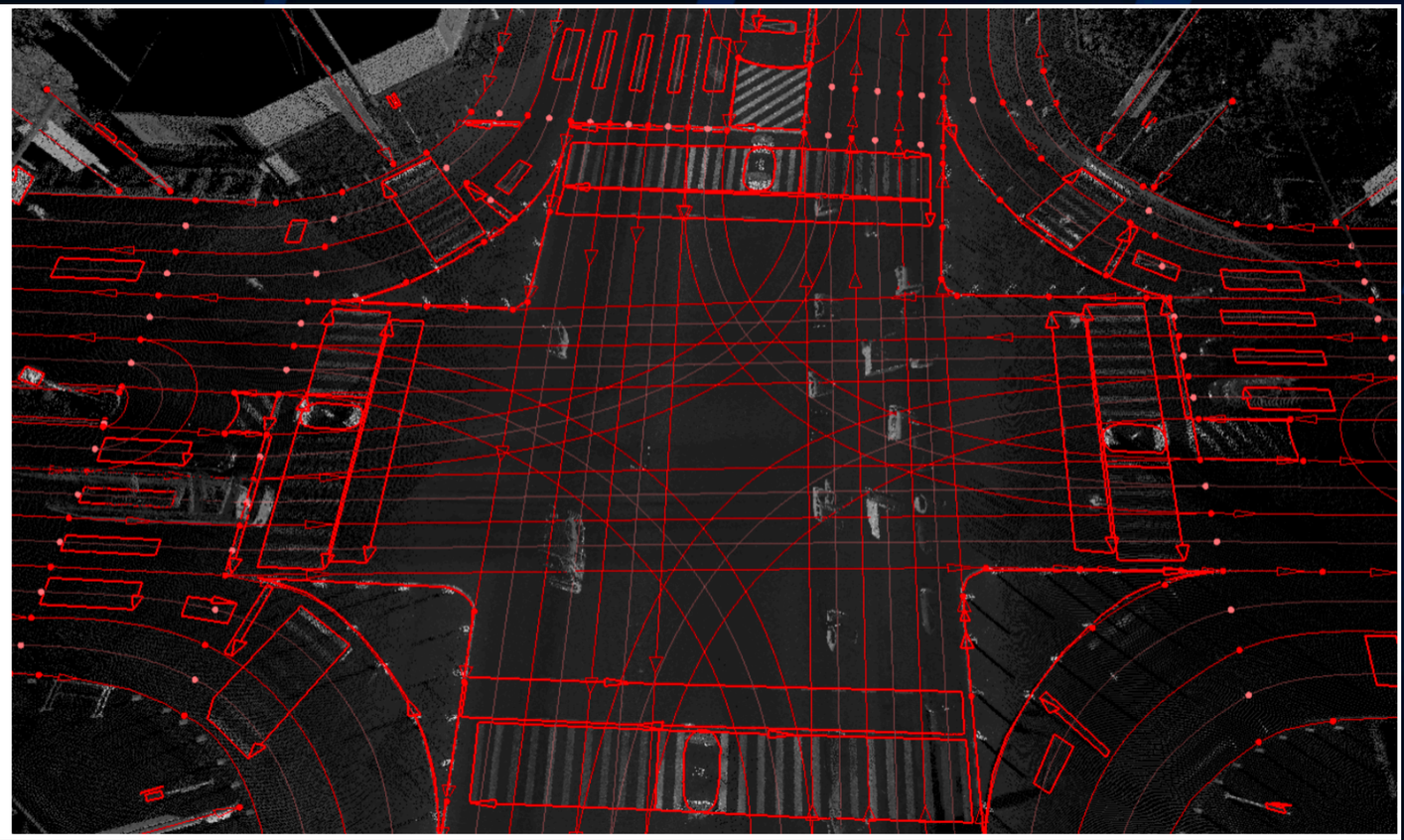




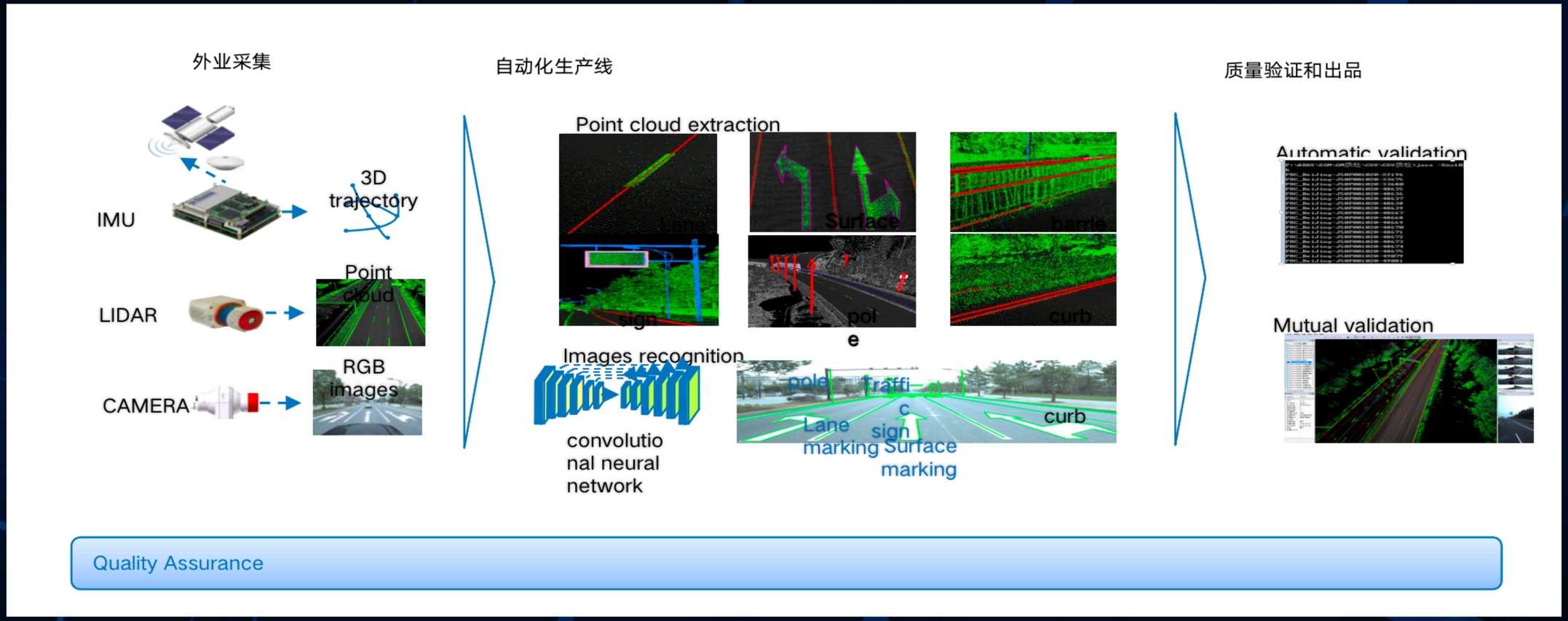
在线编辑



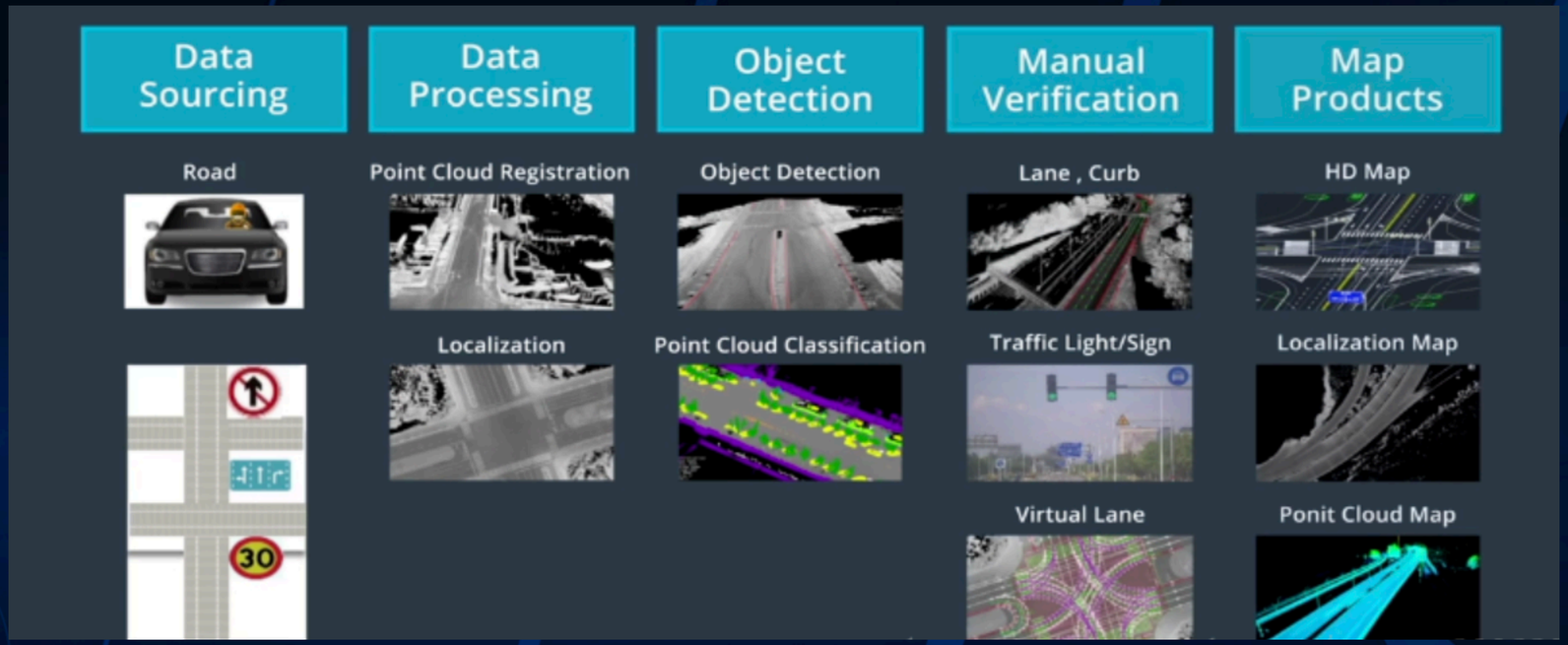




生产流程


外业采集
自动化生产线
质量验证和出品
IMU
3D trajectory
LIDAR
Point cloud
CAMERA
RGB images
Point cloud extraction
Surface
barrie
sign
pol
e
curb
Images recognition
convolutio
nal neural
network
pole
traffi
c
sign
curb
Lane
marking
Surface
marking
Automatic validation
Mutual validation
Quality Assurance

百度Apollo制图流程



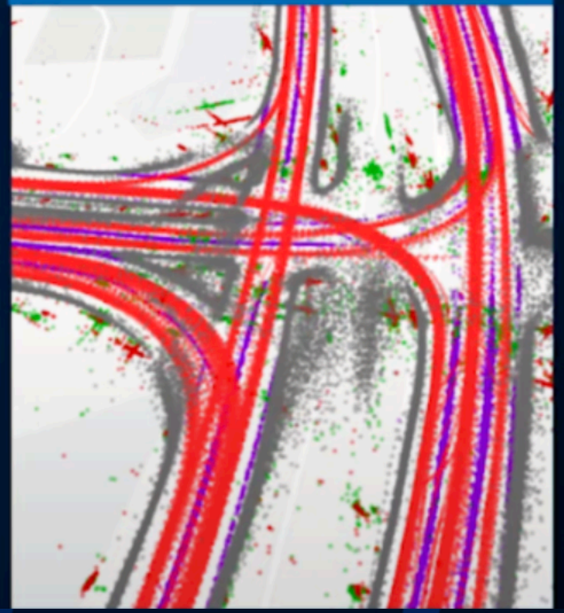
MobileEye制图流程

Harvesting

Single drive reports everything it detected

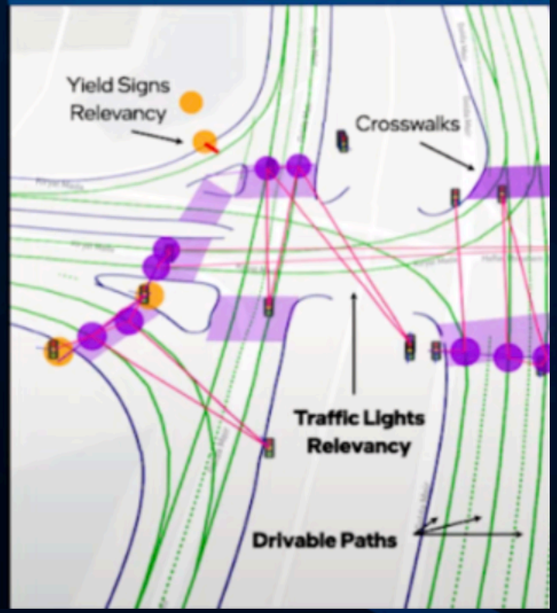
Alignment

Creating a coherent statistical model in a global coordinates system by aligning random crowd drives accurately



Modeling & Semantics

Creating a definitive, actionable modeling of all road elements with accurate semantic description



Harvesting

Road Geometry

- Visual landmarks for localization
- Traffic signs & traffic lights
- Lane marks & road edges
- Crosswalks

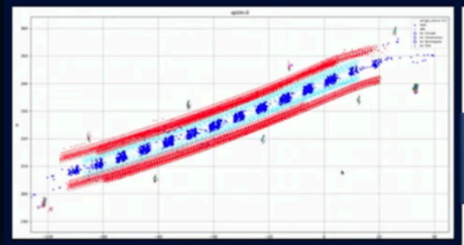
Semantics

- Drivable paths
- TSR type
- Stopping points
- Speed

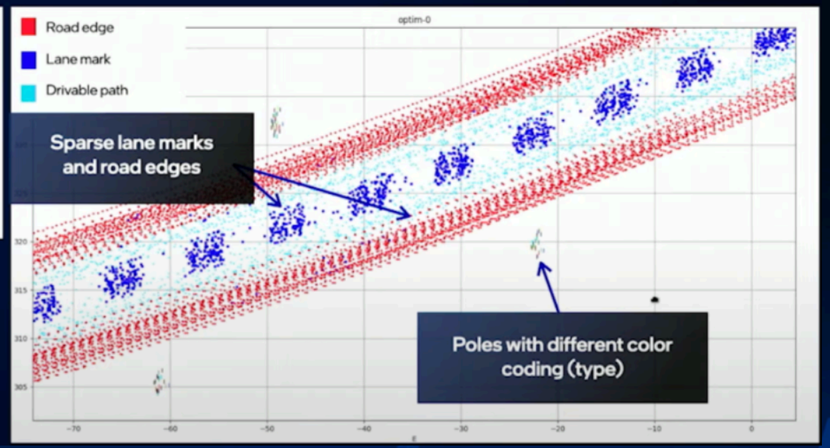
Information is sent as a compact "RSD" (Road Segment Data)
packet, with sparse, economic representation of these elements (10KB/ Km)
Full compliance with GDPR regulation

Aligning

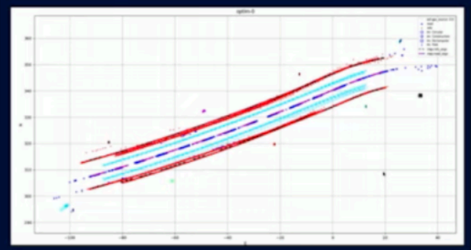
ALIGNMENT BASED ON GPS



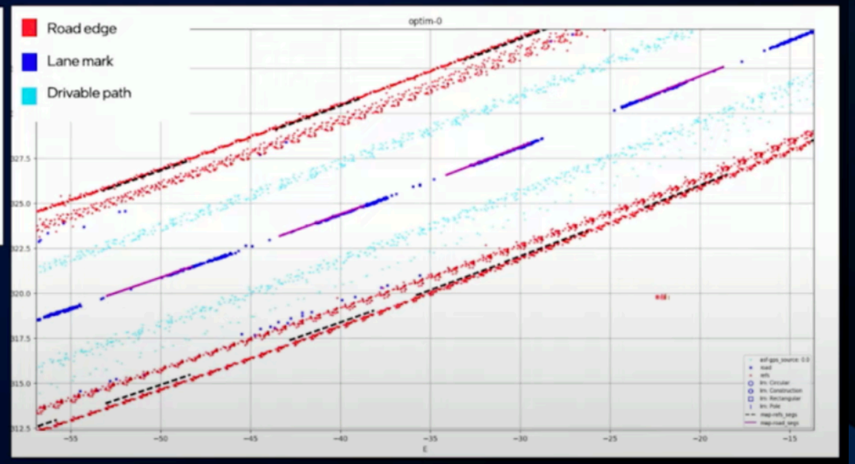
ZOOM IN



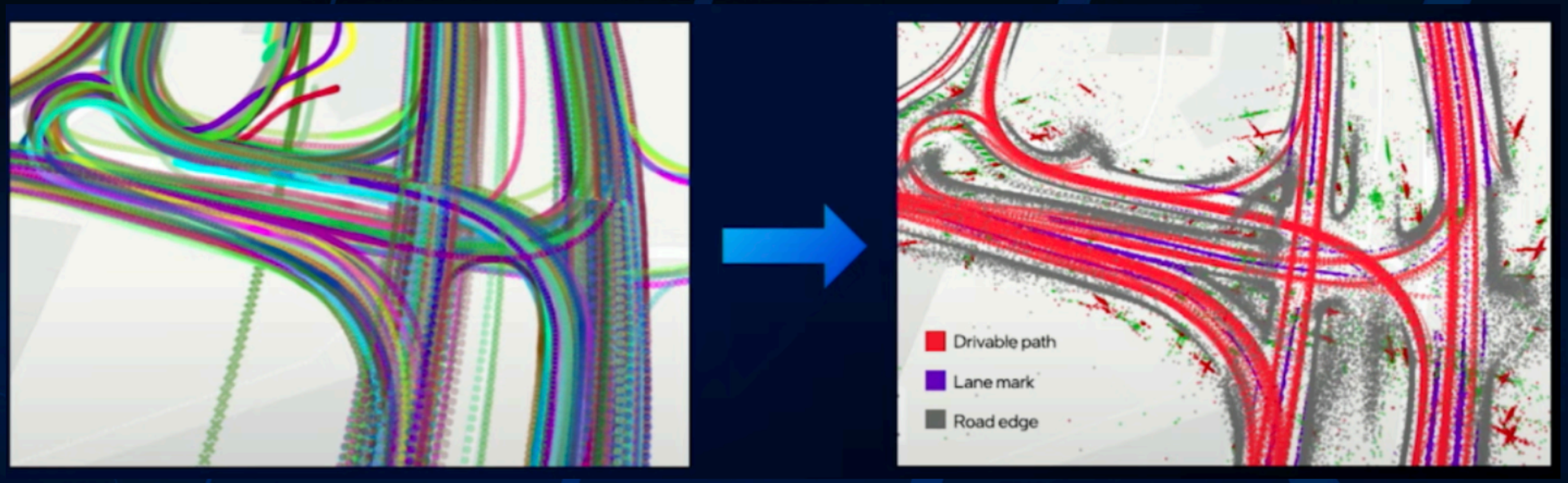
THE RESULT:



ZOOM IN



Aligning

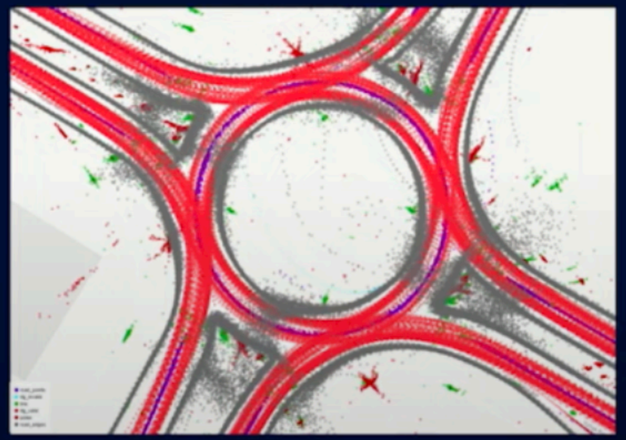


Modeling Process

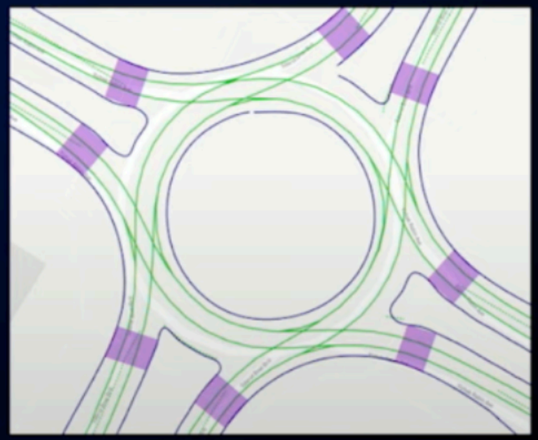
DNN for modelling allows to compensate for noisy data and to increase robustness

Using "handmade" HD maps as a training set

INPUT



OUTPUT



高精地图自动化制作中常用的技术

VSLAM, VO / VISLAM, VIO

	EKF Based	Opt Based
Feature Based	Mono Slam Monoc. FastSlam MSCKF MSCKF2.0	PTAM ORB Slam ORB Slam2.0 Vins Mono ORB Slam3.0
Direct Method	ROVIO	DTAM SVO Vins Fusion DSO

Deep Learning + SLAM

特征提取, 特征匹配

深度估计

位姿估计

重定位

语义信息

CNN SLAM

UnDeepVO

LIFT

VINet

SFM Net

高精地图自动化制作中常用的技术

Lidar+ SLAM

高精地图自动化制作中常用的技术

Deep Learning Perception

见前期专题

高精地图自动化制作中常用的技术

信息融合

多传感器信息最优融合

共性问题

- 数据对准
- 数据关联
- 融合方法
- 传感器不确定度
- Outlier

融合定位与建图:

IMU + GNSS + Wheel Ticks + Steering Angle

IMU + GNSS + Wheel Ticks + Steering Angle + Mono Cam

感知融合:

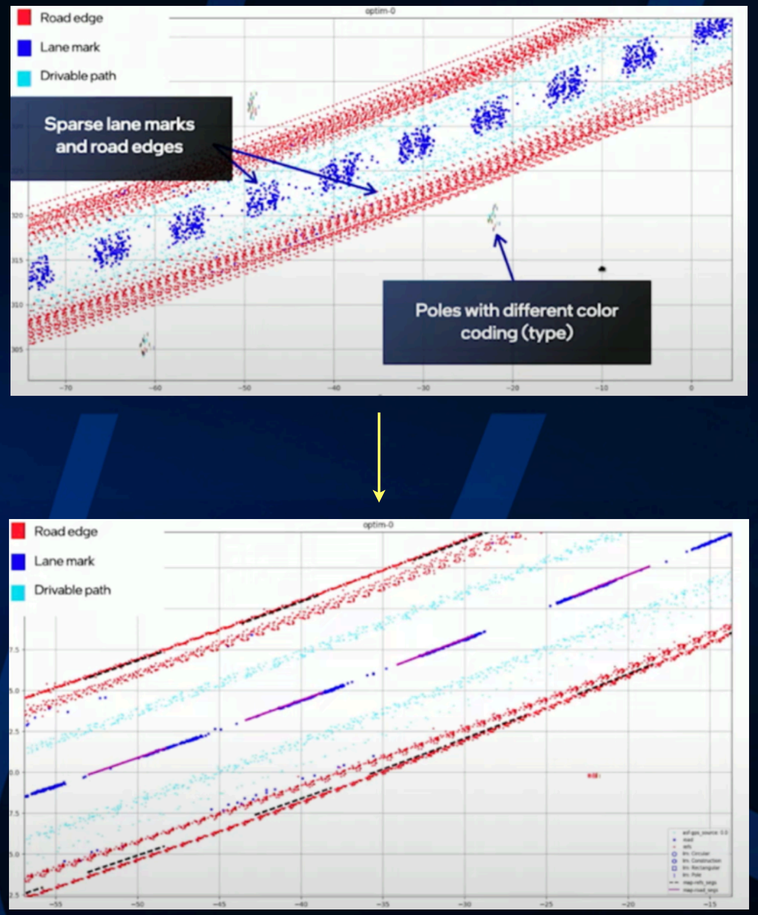
Lidar + Camera

高精地图自动化制作中常用的技术

信息融合

数据级融合

- 基于非线性优化的方法:
  - Geometry Generation
  - PGO Based
  - BA Based
  - Structure Based Opt
- Deep Learning:
  - Matching
  - Geometry Generation
  - Logical Generation
- Deep Learning + 非线性优化:
  - DL Based Matching / Key Points Gen. +
  - Optimization Based Geo Gen. +
  - DL Based Logical Gen.



高精地图自动化制作中部分传感器介绍

MEMS IMU

加速计模型

$$\begin{aligned} \boldsymbol{a}_m &= \boldsymbol{a}_B + \boldsymbol{b}_a + \boldsymbol{n}_a \\ \dot{\boldsymbol{b}}_a &= \boldsymbol{n}_{b_a} \end{aligned}$$

陀螺仪模型

$$\begin{aligned} \boldsymbol{\omega}_m &= \boldsymbol{\omega}_B + \boldsymbol{b}_g + \boldsymbol{n}_g \\ \dot{\boldsymbol{b}}_g &= \boldsymbol{n}_{b_g} \end{aligned}$$

姿态方程

$$\dot{\boldsymbol{R}} = \boldsymbol{R}[\boldsymbol{\omega}_m - \boldsymbol{b}_g - \boldsymbol{n}_g]_\times$$

速度位置方程

$$\begin{aligned} \dot{\boldsymbol{v}} &= \boldsymbol{R}(\boldsymbol{a}_m - \boldsymbol{b}_a - \boldsymbol{n}_a) - \boldsymbol{g} \\ \dot{\boldsymbol{p}} &= \boldsymbol{v} \end{aligned}$$

IMU运动学方程

$$\begin{aligned} \dot{\boldsymbol{R}} &= \boldsymbol{R}[\boldsymbol{\omega}_m - \boldsymbol{b}_g - \boldsymbol{n}_g]_\times \\ \dot{\boldsymbol{v}} &= \boldsymbol{R}(\boldsymbol{a}_m - \boldsymbol{b}_a - \boldsymbol{n}_a) - \boldsymbol{g} \\ \dot{\boldsymbol{p}} &= \boldsymbol{v} \\ \dot{\boldsymbol{b}}_a &= \boldsymbol{n}_{b_a} \\ \dot{\boldsymbol{b}}_g &= \boldsymbol{n}_{b_g} \end{aligned}$$

高精地图自动化制作中部分传感器介绍

MEMS IMU

IMU误差状态运动学方程

$$
\begin{aligned}
\dot{\boldsymbol{\phi}} &= -\hat{\boldsymbol{R}}\delta\boldsymbol{b}_g - \hat{\boldsymbol{R}}\boldsymbol{n}_g \\
\delta\dot{\boldsymbol{v}} &= -[\hat{\boldsymbol{R}}(\boldsymbol{a}_m - \hat{\boldsymbol{b}}_a)]_\times\boldsymbol{\phi} - \hat{\boldsymbol{R}}\delta\boldsymbol{b}_a - \hat{\boldsymbol{R}}\boldsymbol{n}_a \\
\delta\dot{\boldsymbol{p}} &= \delta\boldsymbol{v} \\
\delta\dot{\boldsymbol{b}}_a &= \boldsymbol{n}_{b_a} \\
\delta\dot{\boldsymbol{b}}_g &= \boldsymbol{n}_{b_g}
\end{aligned}
$$

不确定性传播的关键

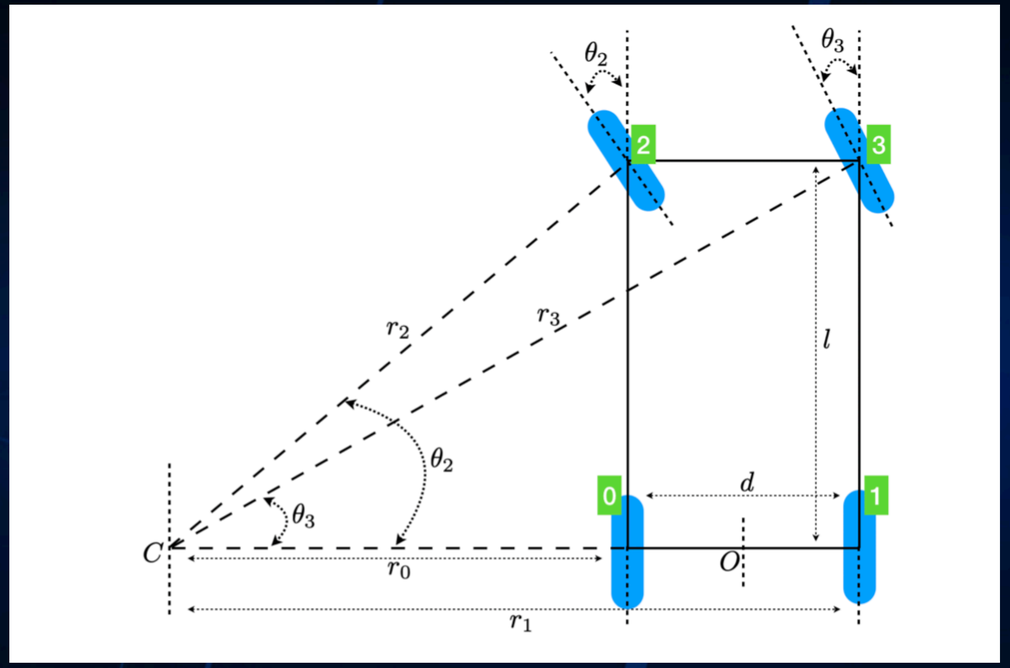
$$
\mathrm{Cov}\left(\int_0^{\Delta t} \boldsymbol{n}\mathrm{d}\tau\right) = \Delta t\mathrm{Cov}(\boldsymbol{n})
$$

累积误差和Bias的关系

$$
\|\delta\boldsymbol{p}(t)\| \propto \|\delta\boldsymbol{b}_g\|t^3, \quad \|\delta\boldsymbol{p}(t)\| \propto \|\delta\boldsymbol{b}_a\|t^2
$$

高精地图自动化制作中部分传感器介绍

Akerman模型



$$\omega = \frac{v_0}{r_0} = \frac{v_1}{r_1} = \frac{v_2}{r_2} = \frac{v_3}{r_3}$$

$$l = r_0 \tan\theta_2 = r_1 \tan\theta_3$$

$$l^2 = r_2^2 - r_0^2 = r_3^2 - r_1^2$$

$$r_1 = r_0 + d$$

高精地图自动化制作中部分传感器介绍

IMU+Wheel Ticks+GNSS松组合车辆定位的例子

基于IMU的车辆状态运动学方程

$$
\begin{aligned}
\boldsymbol{R}_v^w &= \boldsymbol{R}_i^w \boldsymbol{R}_v^i \\
{}^w\boldsymbol{p}_v &= {}^w\boldsymbol{p}_i - \boldsymbol{R}_v^w {}^v\boldsymbol{p}_i \\
{}^w\boldsymbol{v}_v &= {}^w\boldsymbol{v}_i - \boldsymbol{R}_v^w [\boldsymbol{\omega}_v]_\times {}^v\boldsymbol{p}_i \\
\boldsymbol{\omega}_v &= \boldsymbol{R}_i^v (\boldsymbol{\omega} - \boldsymbol{b}_g - \boldsymbol{n}_g) \\
\dot{\boldsymbol{R}}_i^w &= \boldsymbol{R}_i^w [\boldsymbol{\omega} - \boldsymbol{b}_g - \boldsymbol{n}_g]_\times \\
{}^w\dot{\boldsymbol{v}}_i &= \boldsymbol{R}_i^w (\boldsymbol{a} - \boldsymbol{b}_a - \boldsymbol{n}_a) - \boldsymbol{g} \\
{}^w\dot{\boldsymbol{p}}_i &= {}^w\boldsymbol{v}_i \\
\dot{\boldsymbol{R}}_i^v &= \boldsymbol{0} \\
\dot{\alpha} &= 0 \\
\dot{\boldsymbol{b}}_a &= \boldsymbol{n}_{w_a} \\
\dot{\boldsymbol{b}}_g &= \boldsymbol{n}_{w_g}
\end{aligned}
$$

误差状态运动学方程

$$
\begin{aligned}
\dot{\boldsymbol{\phi}}_{i2w} &= -\hat{\boldsymbol{R}}_i^v \delta\boldsymbol{b}_g - \hat{\boldsymbol{R}}_i^v \boldsymbol{n}_g \\
\delta^w\dot{\boldsymbol{v}}_i &= [\hat{\boldsymbol{R}}_i^w (\boldsymbol{a} - \boldsymbol{b}_a)]_\times \boldsymbol{\phi}_{i2w} - \hat{\boldsymbol{R}}_i^w \delta\boldsymbol{b}_a - \hat{\boldsymbol{R}}_i^w \boldsymbol{n}_a \\
\delta^w\dot{\boldsymbol{p}}_i &= \delta^w\boldsymbol{v}_i \\
\delta\dot{\boldsymbol{b}}_a &= \boldsymbol{n}_{w_a} \\
\delta\dot{\boldsymbol{b}}_g &= \boldsymbol{n}_{w_g} \\
\dot{\boldsymbol{\phi}}_{i2v} &= \boldsymbol{0} \\
\delta\dot{\alpha} &= 0
\end{aligned}
$$

高精地图自动化制作中部分传感器介绍

IMU+Wheel Ticks+GNSS松组合车辆定位的例子

重力方向约束

$$\boldsymbol{r} = [0,0,1]^T = \boldsymbol{R}_v^w \boldsymbol{R}_i^v \frac{\boldsymbol{a}_u}{\|\boldsymbol{a}_u\|}$$
$$\boldsymbol{a}_u = \boldsymbol{a} - \boldsymbol{b}_a$$

GNSS约束

$$\boldsymbol{p}_{gps} = {}^w\boldsymbol{p}_v + \boldsymbol{R}_v^{wv}\boldsymbol{p}_g$$
$$\boldsymbol{v}_{gps} = {}^w\boldsymbol{v}_v + \boldsymbol{R}_v^w[\boldsymbol{R}_i^v(\boldsymbol{\omega} - \boldsymbol{b}_g)]_\times {}^v\boldsymbol{p}_g$$

Akerman专项约束

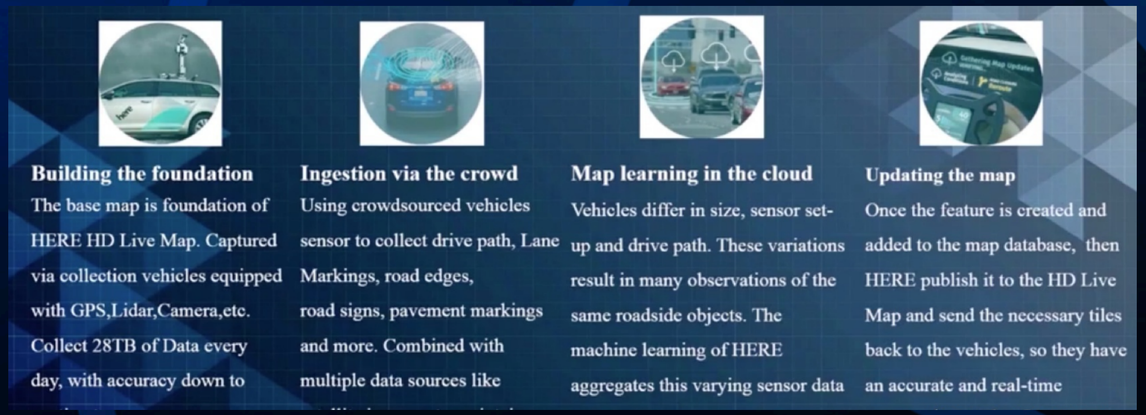
$$\boldsymbol{r} = [0,0,0]^T = \alpha[s,0,0]^T - \boldsymbol{R}_w^v {}^w\boldsymbol{v}_v$$

提 纲

- 自动驾驶是否需要高精地图
- 什么是高精地图
- 高精地图的生产
- 高精地图主要厂商

HERE

- 为个人和公司提供地图和位置数据以及相关服务的
- 该公司由一家德国汽车公司组成的财团持有多数股权，其他公司也持有少数股权
- 它的历史可以追溯到美国诺基亚于2007年收购了总部位于芬兰的Navteq
- 目前总部设在阿姆斯特丹，8000员工

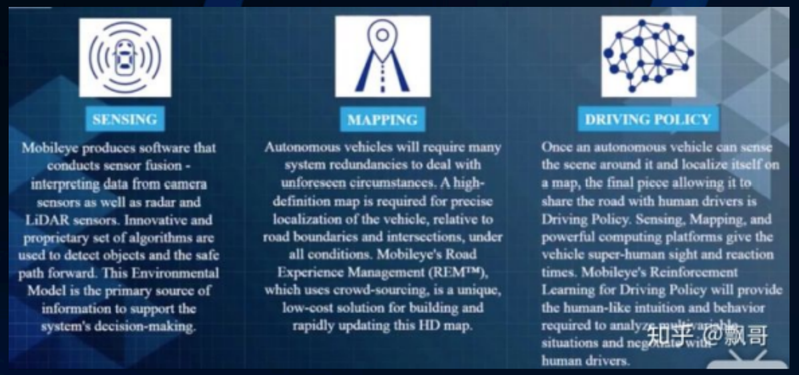


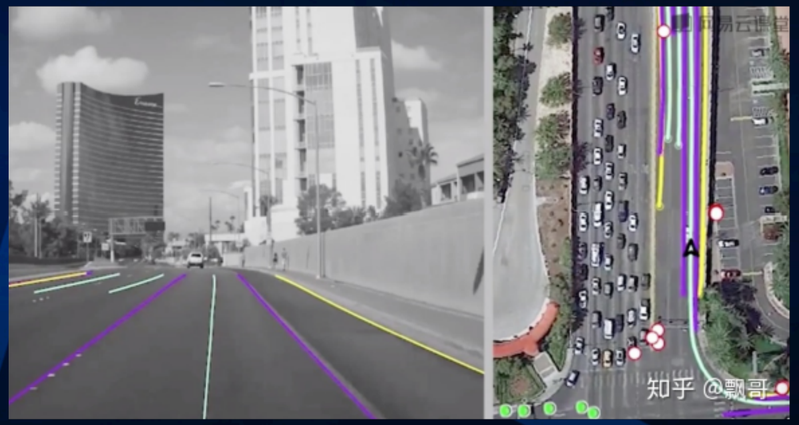
MobileEye

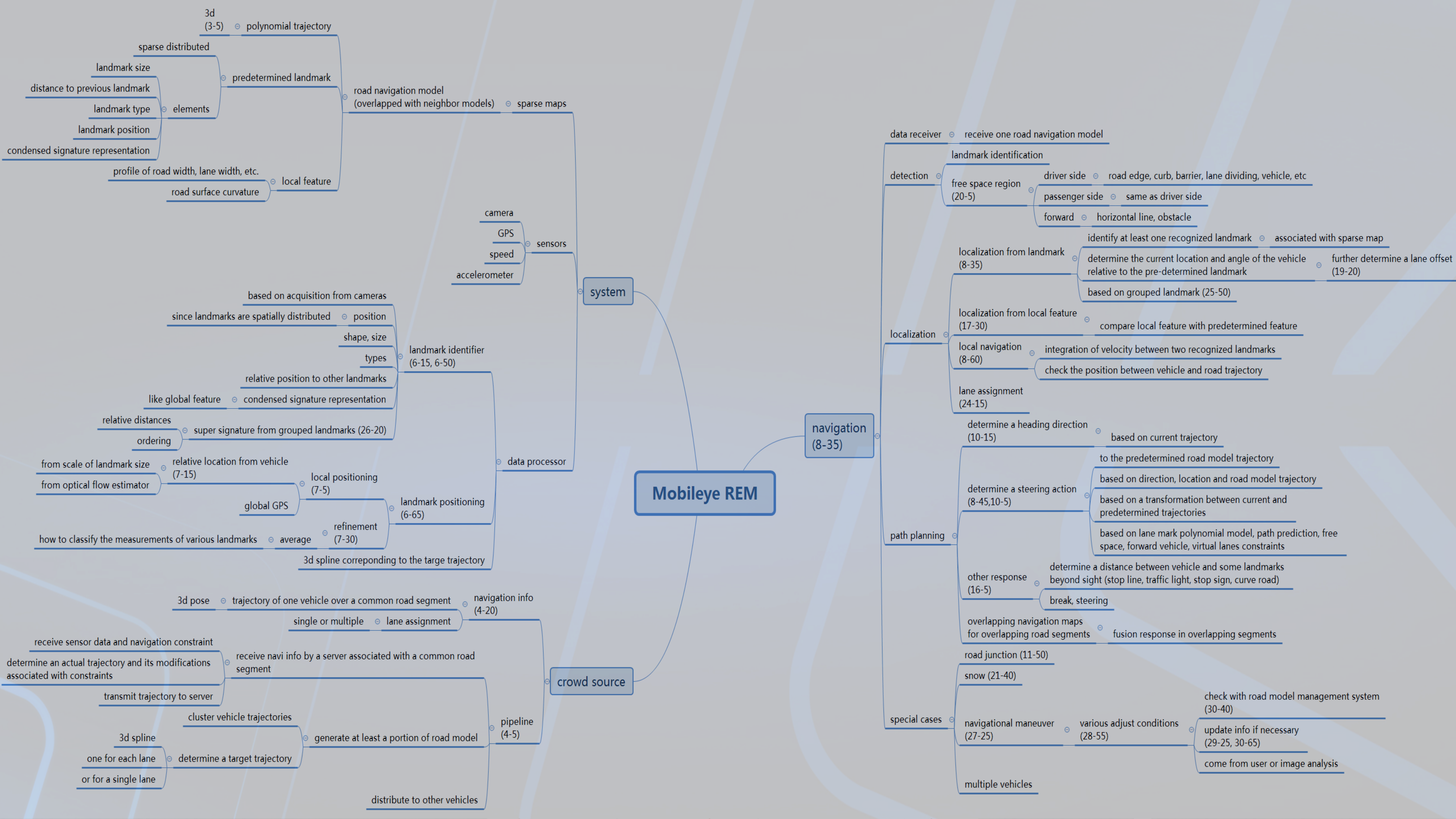
- 以色列公司，Intel子公司

- Mobileye的技术积累非常雄厚
  - 合作了 25家汽车厂商
  - 全世界有 2500 万辆汽车上装载了 Mobileye 系统
  - 13家车厂用他的技术做自动驾驶

- REM(道路经验管理)分三层:
  - 数据收割层（车端）
    - 道路分段数据（RSD）
    - 10KB/公里
  - 地图聚合层（云端）
    - 路书（Road Book）
  - 地图消费层（自动驾驶车辆）
    - 终端车辆定位






Mobileye REM
system
road navigation model
(overlapped with neighbor models)
sparse maps
3d
(3-5)
polynomial trajectory
predetermined landmark
sparse distributed
elements
landmark size
distance to previous landmark
landmark type
landmark position
condensed signature representation
local feature
profile of road width, lane width, etc.
road surface curvature
sensors
camera
GPS
speed
accelerometer
data processor
landmark identifier
(6-15, 6-50)
based on acquisition from cameras
position
since landmarks are spatially distributed
shape, size
types
relative position to other landmarks
condensed signature representation
like global feature
super signature from grouped landmarks (26-20)
relative distances
ordering
local positioning
(7-5)
relative location from vehicle
(7-15)
from scale of landmark size
from optical flow estimator
global GPS
landmark positioning
(6-65)
refinement
(7-30)
average
how to classify the measurements of various landmarks
3d spline correponding to the targe trajectory
crowd source
navigation info
(4-20)
trajectory of one vehicle over a common road segment
3d pose
lane assignment
single or multiple
receive navi info by a server associated with a common road segment
receive sensor data and navigation constraint
determine an actual trajectory and its modifications associated with constraints
transmit trajectory to server
pipeline
(4-5)
generate at least a portion of road model
cluster vehicle trajectories
determine a target trajectory
3d spline
one for each lane
or for a single lane
distribute to other vehicles
navigation
(8-35)
data receiver
receive one road navigation model
detection
landmark identification
free space region
(20-5)
driver side
road edge, curb, barrier, lane dividing, vehicle, etc
passenger side
same as driver side
forward
horizontal line, obstacle
localization
localization from landmark
(8-35)
identify at least one recognized landmark
associated with sparse map
determine the current location and angle of the vehicle relative to the pre-determined landmark
further determine a lane offset
(19-20)
based on grouped landmark (25-50)
localization from local feature
(17-30)
compare local feature with predetermined feature
local navigation
(8-60)
integration of velocity between two recognized landmarks
check the position between vehicle and road trajectory
lane assignment
(24-15)
path planning
determine a heading direction
(10-15)
based on current trajectory
determine a steering action
(8-45,10-5)
to the predetermined road model trajectory
based on direction, location and road model trajectory
based on a transformation between current and predetermined trajectories
based on lane mark polynomial model, path prediction, free space, forward vehicle, virtual lanes constraints
other response
(16-5)
determine a distance between vehicle and some landmarks beyond sight (stop line, traffic light, stop sign, curve road)
break, steering
overlapping navigation maps
for overlapping road segments
fusion response in overlapping segments
special cases
road junction (11-50)
snow (21-40)
navigational maneuver
(27-25)
various adjust conditions
(28-55)
check with road model management system
(30-40)
update info if necessary
(29-25, 30-65)
come from user or image analysis
multiple vehicles

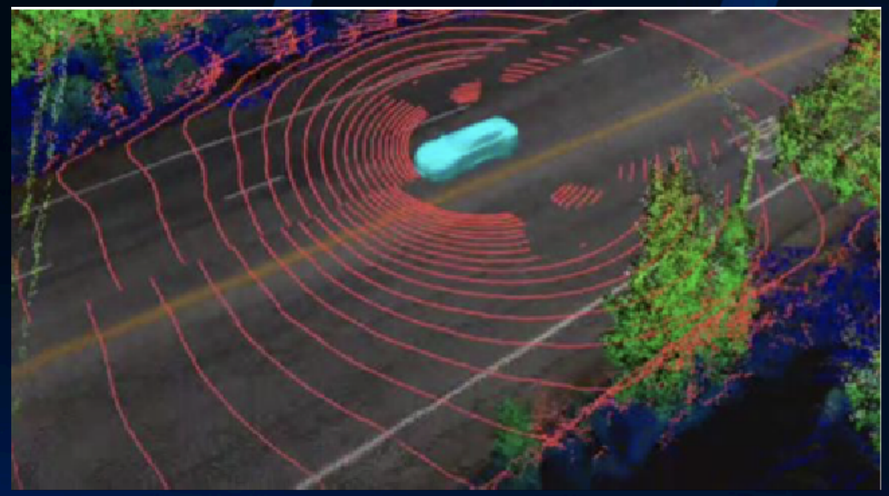
TomTom

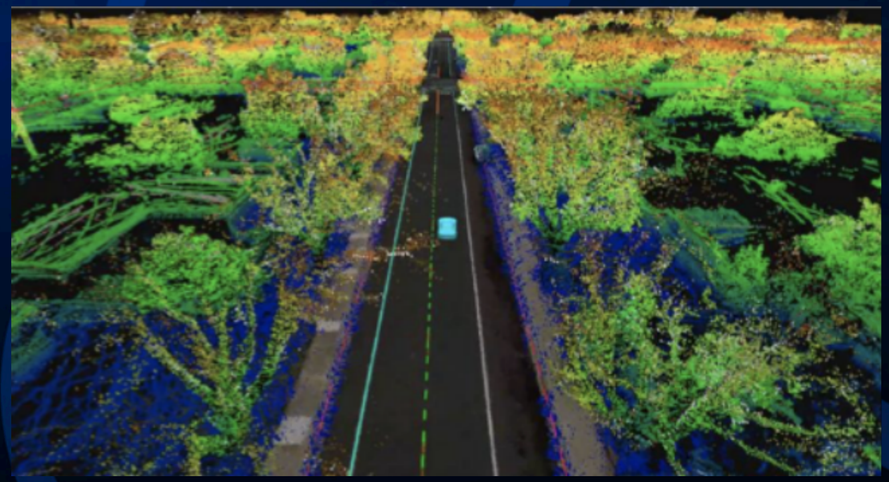
- TomTom是一家主营业务为地图、导航和GPS设备的荷兰公司
  - 总部位于阿姆斯特丹
  - 公司始于1991年，目前约有4000名员工
  - 目前此公司有消费电子、汽车工业、地图授权和Telematics几大业务分支
  - 图商一般都是提供地图加定位的方案

- 定位方案：Tomtom提供的定位方案是基于点云的，它起名叫做road dna，它本身是基于匹配的
  - TomTom HD Map with RoadDNA，是一个具有革命性的、高度精确的数字地图产品，即使在高速行驶中，也可帮助自动驾驶车辆精确定位和计划机动
  - 该地图产品已经可以在美国和欧洲的大部分主要高速公路使用

DeepMap

- 硅谷，华人创业公司
- 激光+视觉
- cm级的HD-MAP精度
- 实时定位能力





Continental

- 德国老牌汽车零件供应商
- eHorizon RoadDB：
  - 车辆侧利用视觉收集道路几何和对象信息，压缩后将矢量数据传输到云端
  - 云端从海量车辆提供的数据生成一个高度准确的道路模型
- eHorizon RoadDB 同时提供精确的车辆定位方案
  - E2E解决方案
  - 包括车载端软件和云服务
  - 建立在现有的汽车技术和机器学习等方法基础上

THANKS FOR YOUR ATTENTION
「通过智能智造，创造更美好的出行生活」